

Introduction to IT Bridge Course

Semester End Examination Solutions (E1PA105B)

Section A: Information & Documentation

1. Classify various sources of information.

Information sources are generally classified into three categories:

- **Primary Sources:** Original materials, such as research reports, diaries, and eyewitness accounts.
- **Secondary Sources:** Interpretations and evaluations of primary sources, like textbooks and articles.
- **Tertiary Sources:** Collections and indexes that organize primary and secondary sources, such as encyclopedias and databases. [cite: 256]

2. Apply document formatting features in MS Word to create a professional report.

To create a professional report, one should use formatting features such as:

- **Styles:** Apply Heading styles for a consistent look and to generate an automatic Table of Contents.
- **Layout:** Adjust margins, orientation, and use page breaks.
- **References:** Insert citations, footnotes, and a bibliography.
- **Insertions:** Use tables, charts, and page numbers for clarity. [cite: 256]

Section B: Systems & Memory Management

3. Analyze paging as a memory management technique.

Paging is a memory management scheme that eliminates the need for contiguous allocation of physical memory. It divides physical memory into fixed-size blocks (frames) and logical memory into blocks of the same size (pages). This technique helps in reducing external fragmentation and allows processes to reside in non-contiguous memory locations. [cite: 256]

4. Evaluate deadlock prevention and avoidance techniques.

Deadlock management includes:

- **Prevention:** Ensuring that at least one of the necessary conditions for deadlock (Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait) never holds.
- **Avoidance:** Using algorithms like the Banker's Algorithm to dynamically check the resource-allocation state to ensure that a circular wait condition can never exist. [cite: 256]

Section C: Graphics & Communications

5 & 6. Graphics Algorithms (Midpoint Circle & Seed Fill).

- **Midpoint Circle Drawing:** An algorithm used to determine the points needed for drawing a circle by calculating the midpoint between pixels to decide which pixel is closer to the circle path. [cite: 256]
- **Seed Fill Algorithm:** A technique used to fill a bounded area with a specific color by starting from an interior "seed" point and coloring neighboring pixels until the boundary is reached. [cite: 256]

7. Analyze the components of a data communication system.

A basic data communication system consists of five components:

- **Message:** The information to be communicated.
- **Sender:** The device that sends the message.
- **Receiver:** The device that receives the message.
- **Transmission Medium:** The physical path by which a message travels.
- **Protocol:** A set of rules that govern data communication. [cite: 258]